Conclusion (Discussion) ~ Active Inference for the Social Sciences 2023

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Hello, welcome everybody. It's November 1st, 2023, and we are in the closing discussion section for this first instance of active inference for the social sciences. It's been a journey. I'm sure we'll have many fun things to talk about and explore today. So, Avil, please take it from here and maybe give an overview of what we're going to do and talk about today.
Avil, Okay, so this is the conclusion session of our course on active inference and for the social science. We have talked extensively about active inference, maybe less extensively about for the social science parts. So the goal of the session would be to assess what we talked about and how it is relevant or not to the practice of social sciences.
Basically, this is a discussion session, so there is no clear plan, but I'd like to address how formally active inference accounts for agency and normativity and identity as a kind of integrated phenomenon, how it can compare or add to our grounds or clash with, I don't know, sociological and political model of agency and of social evolution and of etc. etc. All the topics that we addressed, which are numerous, and discuss about the plausible future for active inference, what it can be used to, maybe what it will be used to, certainly, how it may change and how it's relevant to the discussion.
So, yes, that is the crux of it. We have brought in Alexei, who is a philosopher interested in social ontology, specifically the study of social kinds, which is from Kairos Research. And I do not know who Colin is, but he's with us. Hi, Colin. So how should we proceed, Daniel? What do we do?
If you'd like, maybe pose a specific question or prompt, which could be preceded by a statement, and then we can all hear everyone's voice on that. Okay, so a good question would be, active inference describes agency and normativity and identity as a kind of integrated phenomenon. I think we should go over how precisely it works.
So, yes, let's do that. I will find some, how do you say, illustration that will make it easier.
But let's hear people's thoughts on that, please.
Can you pose your question just a tad more clearly?
How are agency, normativity, and identity addressed separately and together in active inference?
Mal, go for it.

So, I think we like to look at things in terms of interrelations. In active inference, nothing is defined by itself. It's defined in relation to other things. And this definition in relation to other things also defines the types of constraints that will be pushed on the thing which is separate from other things. Once you have this, you necessitate these interrelationships to be formative, ontological, and fundamentally contextual. So, nothing, no relationship is perfectly reproducible across a different context. It will take on a hermeneutics, a semantics, a different context, given the exact context of the spatial temporality of the entities.

Those are some first thoughts. Does anybody have, like, big objections to what I just said?

Okay, so let's start.

Oh, yeah, please continue. Sorry.

Let's start with the concept of agency, for instance. Agency is just action selection, right?

So, agents are modeled. They have a generative model, and they use it to predict, like, sensory inputs based on possible actions. They try to select those to minimize the difference between predicted and sensory inputs. So, they're just trying to minimize free energy. And this minimization is fundamentally due to the relationship of constraints in their environment. So, let's imagine that I am an individual who...

Oh, sorry. I saw a comment, and I didn't...

So, yeah. So, you have an individual that tries to exert control over the environment. Now, if you scale this in nested structures, you can try to think of power structures and how agency within those power structures is modulated by the different kinds of constraints that are both embedded within the environment itself, within the materiality of the environment, and also within the normative aspects of the nested structures, which pull and push an individual within certain paths through some kinds of deontic cues. So, that's how you get to the notions of encoding agents' beliefs and preferences about the world within the generative model through a form of normativity, which is just a higher precision over a specific type of outcome or mapping. So, your prior beliefs here come from different layers in the hierarchy, and these act as what we call culture or normativity. I could stop here if you want to say something, Avel.

So, I agree with what you said at a high level, conceptual scale, but I think it's important to be clear about what formally governs the statement, because what we have formally is much more limited than what we agree the interpretation could be. So, can you see the screen I'm sharing?

Mm-hmm.

Can you see the mouse doing aids?

Yeah.

Okay. So, at core, Active Inference is a framework that provides geometry for GEMICOL systems. That is not a very straightforward way to put it, but what we have is a system with a space of states, and we have a flow over the system. A flow is a function that basically tells you where a specific point goes with time. And what Active Inference says is that given specific patterns of GEMICOL coupling and of statistical dependencies, we have agent-like behavior. So, the simplest kind of of the basic requirement of Active Inference, sorry, differential principle, which is a core motivation for Active Inference is that you have an identical system, so something with states and a flow and some noise, because we need noise to get information. And it has a boundary in the middle of it that statistically separates the external internal states. By statistically separate, I mean that every information about what's going on there that is in there disappears when condition on there. So, the information I have about this that I could read here, I can also read there. And that's subtly different from an identical coupling, but this is formalism. Let's not get too much into this. And then you have more

complex behavior when you can partition sensory inactive states and your particle can do things.

So, in this specific context, basically we can write a geometry for the fluctuation of the system.

And this geometry entails belief of original states. So, this internal state knows things in a sense about

what's going on out there through its coupling, through sensory inactive states. And the flow it has

internally, the way its internal states move, they entail

a displacement of their belief. And this displacement happens to be near optimal given Bayes constraints, blah, blah, blah.

And the real interesting stuff happens in the case of strange particles, which are particles where,

as you can see, there is no direct effect from actions, active states, to internal states. The only action you have is mediated by sensory and external states.

In this case, the active states themselves, so the policy of an agent, it becomes the subject of beliefs of the agent.

If it becomes the subject of beliefs, you can have statistical inference. This is the crux of the framework.

And if you can have statistical inference, you can have quite complex, multi-level statistical inference over your actions.

And those are basically what we call decision taking or what we call planning.

So, what we have here is a formal architecture that can derive the kind of complicated agency,

semantic semantic, not only agency, but agency that is embedded in a specific conceptual framework that is

relative to the specific structure of an agent and their environment and the coupling in between.

All that is a traceback, basically, to dynamics. So, we have a proof of existence that in the world,

that everything is dynamics. We can have agency, we can have complex decision making, and other form of inference over one's own actions.

And then you have complications to apply directly, of course, because the world is likely not in its barest form.

It doesn't have a list of states that are...

Okay. There is not a list of states that are possible, and that's it.

And there is one coupling that takes those states themselves, and nothing can change ever.

This is not likely to be the case.

But we have a minimal model of agency that is purely physicalist, that is purely based on the

causal relationship in the most minimal way we can possibly represent it.

And this is quite huge.

And this tells us... This gives us a very, very strong prior over what we can...

What is the stuff that agency is made of? And the stuff that agency is made of, at least higher order agency that we like

think about here in the social context, is belief over oneself.

And so, to control actions is to control belief of oneself. And this is this...

This duality, this...

Like, dual aspect of statistical organization as a complex cognitive agent.

That is, in my opinion, at the moment, given the capabilities of modeling and formalism we have,

the core contribution of active inference to social sciences, written on large,
is in psychology, neuroscience, like, science of social stuff.
And this gives you a very strong prior over what agency is made of and the kind of stuff that can exist out there in the social world.
And the kind of dynamics that bring about its stuff.
So, but there is a quite big bridge to be made between the pure dynamical system for formalism,
in fact, in France, and the kind of open-ended evolution that happens out there.
And I do think there is a high level of precaution to be taken when we talk about this kind of stuff.
And, yes.
Does anyone want to react or agree or disagree violently?
Because I just attacked the common creeds.
I'm not sure I perceived an attack, perhaps.
From what I understood of what you said, basically,
you have information geometry.
It formalizes a relief space in a sort of geometric manifold.
Again, we see this kind of ail
knows it has available. And this becomes normative because of the strength of the attractors
within the belief space, which become the geometric landscape basically. So we can sort
of model the attraction through the curvature of the space around the attractor. And so
high curvature means a strong normative force, which pulls the agent to adopt a certain, I like to say
social scripts because that's what I've worked on, but you know, any sort of policy and or phenomenology
norm. And so in this context, the agent's agency is kind of complicated because it's both the ability
to sort of navigate the landscape, to move closer or farther away from the attractors. And so the
degree to which an agent has agency is a degree to which it doesn't go necessarily directly towards
the attractor, but has some degree of reflexivity over its own action space in order to navigate
that, the different kinds of attractors that may be available to it. And so we could see that the
momentum would be like affected by personal experiences, education, innate tendencies, or even,
you know, the tendency to resist or yield to social pressures. I could go on, but I know that A, you have
your hand up. So why don't you go?
Okay, I guess Avel could answer before I talk a bit. So I was just trying to have my hand raised before
talking. So if Avel wants to answer you and afterwards I can talk.
I'm okay. I don't have anything specific to answer.
Okay. Okay. So you talked about the dual aspect of statistical organization of complex agents.
And you said that it was a core contribution of active influence to social sciences.

Can you dig a bit on that? And yeah, just articulate it a bit more so we could grasp it better.

And then I could ask questions.

So do I still share my screen? Yes. So basically what we have in here is a formalism that builds from very basic dynamics in the like just

we have a space and there is stuff that depends on the space. Zero level physical presentation of literally anything can go there.

And from there we build a model of first self organization. Not quite that.

Agentive self organization, let's say something that can build a specific stable ecology and maintain itself in this ecology.

That is what we have here. And then given the lack of coupling between active and internal states, we can build a

essentially more complex stuff that can implement higher order beliefs about themselves and what they do.

And like do decision making that is conditioned on information they have about themselves and like planification under the form of beliefs over their own action.

And as such, this is like, this is just math. This says nothing about the world in and of itself.

Until we claim it corresponds to things in the world. And we can claim that this is a basic conceptual model that maps well into

enculturation, basically social organization engagement with the social cultural world.

Because then we can account of the human ability for very, very, very wide open-ended learning as something that comes from there,

as something that allows us to infer things about ourselves and then develop complex policies and like them, blah, blah, blah.

And because to my knowledge, there is no competing model that accounts for anything near that, like the only competition, like the competition, I shouldn't talk of competition, but the models of cognitive organization and

culturation that exist that exist that are trying to run themselves in physics, like a chemical system, they are just talking about dynamics.

They are specifically avoiding talking about anything like content or semantics.

So, de facto, this piece of math has a monopoly on building semantics from the ground.

And what it tells you is that agency and identity and relativity, it's all the same phenomenon.

It's conditioned by self-belief, basically. That is constructed in this specific way.

And we can, how to say, because this is the only way that we can account for higher order agency, then we have, at least I would argue, we have, we should have a strong

priors for theories of agency, either mechanical or conceptual, that vibe well with that picture.

So, this is the climb I'm making. I guess I want to do too much detail, so I hope this is not.

Okay. Okay, so I have some, maybe not questions, but some points to make now. Thank you.

So, from what I see here, we could put the active inference paradigm into the individualistic methodological individualism

framework in sociology. So, the idea is that you try to model an individual and then, when you have the good way to model the individual, you can generalize to what everyone is doing.

So, for instance, if I'm working on something in certain way, then I could generalize about that and say, okay, sociologically, I have this constraint to work on this stuff.

And then, not like everyone is doing that, but it could go like that. But what is interesting here is that it's not only about one individual, but also that, for instance, let's, let's take the external state.

You could say that external state here or other agents and also that, okay, so you have the active state in the internal state. So, we could say that your internal states are totally influenced by another agent.

So, we could do something like that with two individuals, I guess. So, one is one of the agents here and the other drives the internal states of the model because they are really influential on yourself.

So, there are some limitations about this kind of way of thinking because, in general, in sociology or in sociological sciences, what you would like to say is that there are some agents at a higher level than only individuals such as you and I.

So, the trick could be to say that some states, for instance, or some firms are agents for and by themselves. So, you have agents which are not individuals or persons.

So, they are known to make states, they are known to make states

about how we could have, let's say, two active agents trying to talk together.

Yeah, we have such simulations that already exist. Over various kinds of scales, we have two agents

coordinating that are animals, like birds singing and over time coordinating over the song they're going to sing.

We have several models of leader-follower where you have one agent, two agents starting from relatively the same point, but one agent developing more confidence and therefore developing a dynamic of leader-follower with the other agent.

This is work that's possibly going to be followed by Francesco Balzan, who is working on models of education and how maybe the teacher and student dynamic may be cast under these terms.

I've done work with Axel Constant and Maxwell Ramstad in terms of active inference in social scripts.

That's what Ava was mentioning. And this notion that through the cues of the environment, your own model is shaped both by the environment, which includes other people, and the way in which you self-evidence within this environment.

So for instance, there's two ways we can understand your question first.

We can say that minimizing free energy is basically an agent that has a form of autonomy and it's constantly updating its model to align with the environment.

So by virtue of doing this, it's not fundamentally individual, but each agent has its own model of the world. Each agent reflects a perspective, a very unique moment in space-time that will incur some errors relative to others.

So it's not that they necessarily interpreted things wrong, although they might have, but because their perspective is slightly off from somebody else, there's necessarily a degree of incommunicability in this sense.

So here self-evidencing really entails that identity, you create a sort of identity here through the fact that you may have different preferred states from somebody else based on your values, beliefs, your social conditioning.

But again, these beliefs and values and social conditioning have been shaped. And again, we're talking about these trouble of attractors by other people.

So we try more and more to challenge this notion of individualism by emphasizing the relational aspect of identity, which means that identity is not pre-established, but continuously formed through the interaction with other agents.

And again, if we go back to information geometry, the geometry of the information space influences how agents perceive their paths towards minimizing free energy and affect their choices, which again, give you the possibility to understand how agents might be very strongly pushed by other agents.

And let's imagine that has more power than you and is trying to influence you to do something, someone that has very high charisma or someone who has the power to say destroy your life.

You are more likely to shape your beliefs to align with.

And therefore, we can understand individualism under a very, very different light here.

Avalda, do you want to add something?

Yeah, we want to add to this, we want

the importance of social scripts appears to be greater than the influence of individual upon individual.

And that's interesting because it's not something that we would tend to think as individuals,

that we are integrating the script.

In fact, the whole idea of seeing a script as something which is social

was something that I find that probably most people would find a rather strange idea off the top of our heads.

So, yeah, is that a fair assessment of what you're trying to say?

And the other thing I wanted to ask about was a few times in talks recently and here today,

I've heard people talking about identity and the active inference view of identity.

I'm not sure that I really understand what identity means within an active inference framework

as distinct from just our internal model of who we are.

Is there an other idea that we're thinking of here?

And more particularly, the word identity has a lot of baggage and usage in current and contemporary society.

Are we using it in a different way?

And if so, how is it different?

And potentially, how does that give us insight and the ability to engage with the ongoing discussions in society

in a potentially fruitful way?

So, I think I lost you for the last part of your question,

which seemed to be about the difference between our conception of identity

and the conception generally used, if that's correct?

You're muted, Colin.

Sorry, yeah, I was referring to the concept of identity as used in society.

Right, so I mean, I'm not going to speak to every use of the term identity

because it's a very, very useful term that's used in law, that's used in physics,

it's used in sociology and psychology, so it's used a lot.

All I can do is explain roughly how we consider identity formation

and then if you have a specific definition in mind for a specific field,

we can maybe tackle that one.

Does that sound okay?

Yeah, to be clear, I was thinking specifically about identity

as determined by group membership in sociology.

Okay, so I'm going to tackle social scripts first.

Yeah.

I'm going to tackle social scripts first and then I'm going to come back to identity

because I think it fits.

I've worked on that as well.

I know that Daniel and Avril have some work that relates to this as well.

But so your first question was on the importance of social scripts

and what it means for cognition, if I understood correctly.

So from an internalist perspective,

social scripts are cognitive structures.

They guide behaviors.

They're acting like brackets.

We use the word constraints a lot.

It's not a perfect guide because no semantic marker is a perfect guide.

Everything is up for some degree of interpretation.

It's just that there are some interpretations that are so strongly linked to some degree of embodiment

that they sort of anchor the rest of our semantics.

So for instance, I can't stop breathing.
If I wanted to, I can't.
I will stop, but eventually my body will restart on its own.
And so there's such kinds of natural laws here that create these bottom layer guidelines
that can then be sort of derived to infer other things.
And social scripts are just representational social extensions of these constraints
given a historical context.
So they model and...
Did mouth drop or did I drop?
I guess it's...
They're given to you.
Like somebody will actually tell you what to do and show it to you.
And sometimes they won't be.
Sometimes they're just things you observe.
Sometimes there's things you infer
because you made a connection between a weak script and another weak script.
So if you see something that resembles a chair in a room,
you infer that you have the affordance to sit on it.
So that's a form of script.
And then you see other people doing the same and the script sort of cements.
And so now you associate this to another part of a sequence.
And this becomes something that allows you to temporalize and basically leverage your B matrix
in order to exist in the world in a way that doesn't cause friction with other people.
This also gives you action triggers.
There are things you see in the world that now tell you this is the thing you should do now.
So there's this sort of recursive loop that's very inactive that pulls you in a direction.
So the attractor isn't just a state.
It's a path.
You're going to go down a path.
So for now, does that answer the first question?
Yes, thanks.
That's helpful.
Okay.
Go ahead.
Thank you.
To get back on the, like you talked about an active loop,
and I think that's a critical thing to analyze,
which is that the engagement with the world is a cyclic process,

it's a circular process.
It constructs its own conditions.
And so there is no measure of power of structure versus power of agents
because the structure is embedded in the agents,
and the agents are embedded in the structure.
So I'd like to share my screen again
to give some detail over what this entails.
Do I, can I?
Yes, good window.
You should see a PDF with a tiny man,
a person, a mountain, and loops.
Is it okay?
It's okay.
Okay.
Yes.
I see.
Yes.
So this is a presentation I did recently
about the Unification Church,
which is like when you want to be published,
you say new religious movement,
but you can say cult.
It works.
Which originated, I think, in Japan,
and is now spread between Korea, Japan, and the States.
Mostly, we don't have numbers,
but it seems to be the places where it's most implanted.
It was very, very, very remarked
because of the personality cult
around the Reverend Moon, the founder,
and the very overt,
extractive practices of the cult,
like asking you for the equivalent
of 10 years of salary at the time
to take your ancestors from hell
and put them in paradise
by giving money to Reverend Moon specifically.
And they were the topic of a very aggressive

and excessive anti-cult movement in the 70s, 80s.
And now they are becoming a bit noticed again by Vice.
Not only Vice, but notably Vice,
because you have very strange things
like mass MAGA weddings
with people waving rifle.
And that is a very strange organization.
And at the time,
you had a very moral panic,
we say now, around the cult.
You had basically a model of influence
that was that evil people would brainwash you
and then you were commanded by the cult
and or Soviet secret agents from afar.
And they could like say to you,
a horse and this triggered
homocidal mania in which you blah, blah, blah.
So things that are unnaturalistic, I will say.
And notably in reaction to the moral panic,
you had a movement by the religious of religion
that would try to understand
how exactly the cult worked.
And the most famous sociologist to do that
was Aileen Barker.
If I remember well,
she warned the Moonies she was coming,
so you can question her methodology.
But she seems to have done the most thorough
and well-received analysis of them.
So it's quite an important thing.
And she questions that there is such a thing
as brainwashing.
There is a construction of a specific social context
that you have to comply to if you want to come,
but they can force you.
They do not have the capability
to like torture you
and make you into a RC radio commending thing.

That does not work.
And so I forgot the name of the...
So David Frank Taylor explains the influence of the church
in terms of choreography of total participation.
So basically, you have a very huge,
very emotionally investing ritual
that you just have to take part in
to be a member of the community.
And if you are an member of the community,
you will be helped,
you will be loved,
you will be supported,
you will be constantly encouraged
in a slightly coercive way
to invest more of your time and money
in whatever organization.
And if you don't,
then all the support just leaves
because you're a traitor.
You have treason of the true parents
who was kindly out of his own arms
trying to save you like Jesus did,
but better because Jesus failed
by not having children.
That is a very dramatic thing to say
and that is what they said.
And because of the way the organization is set up,
you cannot possibly question that.
Your questioning is not accepted.
There is no private time
in which you can privately question yourself.
You are always engaged with people
who are either formally or informally,
depending on your seniority,
watching you.
And so you have basically a context
that makes it very, very, very easy
to align with the cult values.

And it just happens that those values
they entail self-sacrifice
for the organization,
but also for new members
and total deference to the cult
and its leadership
and anything they tell you to do.
So you know,
it's not kidnapping,
but it's also not exactly free consent
to a specific context.
And from a purely,
I don't know what to say,
mechanical perspective,
they can make you agree
when you are there,
but they cannot make you go there.
So a question is
how precisely
they can, you know,
elicit compliance
away from the direct context
of the choreography of participation.
And there is a,
like, simple answer in my sense,
which is that
we have a coherent sense of identity.
We can't,
we can't only think of ourselves
as a good money,
one minute,
and then step out of the meeting
and not think of ourselves
as a good money.
That doesn't make sense
either we are
or we are not.
At least most people

tend to have
one integrated sense
of themselves
as someone
who is embedded
in a specific social context
and has specific norms
and bears in such a way.
And so the coherence
of self-identity,
it provides a key coupling
between your public
and your private beliefs,
what you are expected
to do in a social context
and what you will effectively do
and perceive the world as privately.
And this,
I would say,
an example of the Moonies,
but in other historical examples
like the emergence of nationalism,
it is a very straightforward confirmation
that indeed coherence
of self-identity,
like the coherence
between what you do
and what you think
of yourself as doing,
is a very important
process
in the structure
of societies,
generally speaking,
at least contemporary societies.
Maybe it was not
the same before.
And so I do not think

like the context
in which it's mobiliar
is of course it changes,
but the mechanism,
I don't think it's new.
And I'd like to get back
on Alex's point
that,
sorry,
A's point,
that the approach
we have is
individualistic.
I do not think
it is the case
because if we are correct,
there is a landscape
of social cultural
affordances,
constraints,
things,
world out there,
and we engage with it.
And there are mechanisms
that makes this landscape
not necessarily identical,
but very coherent
between individuals.
And those mechanisms
are language
and the material symbolism,
developing ritual,
which is also symbolism,
also meaningful,
but maybe less than language.
I don't know.
I just mean that
the mechanism is the same.

And the interculturally
robust practice
of role-taking
and shared essentialities
are trying to do stuff together
given a specific
model of the attribution
of tasks
within the shared goal.
And so these are
human invariants
and all
would be pretty good candidate
to force the integration
of a common model
of material
and structural niche.
And so I think
that there is
a pretty coherent
social material niche
and I think that
we can,
we all engage with it
and we all modify it
and we maybe
do not perceive it
as the same,
but there is still
a strong enough constraints
on the coherence
of this landscape
between individuals
for it to make sense
as a specific entity.
and then
the proper
explanatory scale

of what we do
is not the individual minds,
it is landscape
that we happen to experience
because of our individual minds.
So by the
Plato Academia
of Stanford
on the holism,
basically holism
is a theory
where you agree
that there are
explanations
that are not mediated
by individual states.
and I say
there are explanations
that are not mediated
by individual states
and not only that,
but I can point
to organizations
that control people
while being absolutely
not worried
by individual states,
only creating
a certain context
and then allowing them
to engage with it.
And that would be
compelling in my opinion
to say that the model
we have is maybe
not holist,
maybe a secret
third thing

that is multi-scale,
but certainly
not individualist,
at least in the sense
that you are mobilizing.
So thank you.
I talked more
than I wanted.
So Mao,
you were raising your hands.
I am,
but I think Lorena
wants to go.
I will go after Lorena.
Hey guys.
Yeah,
I think that goes
like directly
into the idea
that coming
a little bit,
coming back
a little bit
to the point
that identity
that we were talking
before,
and if you think
about the model
that we have,
we have external,
the external states
and different agents.
They are always
actually very local
and situated
in the same,
in the same local

constraints
and local environment,
right?
So your external agent
is going to be part,
the other agents
are going to be part
of your external states
and you are part
of theirs,
right?
So the idea
of like laying
down a path
and creating
a specific type
of dynamic
that reacts
to the same
type of input
that you are receiving
and driving
these dynamics
will generate
this sort
of embeddedness,
right?
Because you do
create,
you do have
the same,
or no,
it's not necessarily
the same,
but they're very,
very close to each other,
they're very approximate,
right?

So the idea
of a dynamic here,
if you think
about describing
that in terms
of a laplace
and like in terms
of the dynamics,
you do have
very different
like Lauren's
butterflies,
for instance,
each one goes
through the same
type of route.
However,
they're not identical,
but they're very similar.
They're doing
something very similar.
And that similarity
sort of constrains
again the construction
of that environment.
So if you think
about symbols
and roles
and types
of environment
that you find,
they're going to be
as concrete
as something else
that you find
in your environment,
constraining
and pushing you

towards the same direction
or at least
a very similar,
you can describe
dynamically
in a very similar
type of geometry
if you like.
or a very similar
or this attractor
will drive
a very similar
type of path.
We'll describe
this similar
type of path.
So,
which goes
exactly to the,
Avel,
can you go back
to that
other,
yeah,
the other figure
that you have,
the one
with the environment
as well.
Made by,
yeah,
that one.
So,
exactly that
creates
like a dynamic
that they can
be then forgotten

like in a way
that will be
described
through a specific
mark of blank
that will develop
agents
together.
We push them
constraining,
they're going
to be constrained
together
and they're going
to have a
constructive
kind of
role
in this
embedded,
in this very
embedded
normativity.
But,
yeah,
you go.
Thanks.
Yeah,
so I think
there's a nice
way to tie
all this
as well
to the
internal nature
that we seem
to be
dispelling

but we're not.
We have to talk
about specifically
in what ways
internality
is cast
here
and
I like
to think
of the
effect of
self-esteem
here.
Self-esteem
to me
is the
degree
of fitness
relative
to your
perception
of how
well
you're
managing
to exist
within
these
constraints
and interpret
them in a
way that
gets other
people
to validate
and minimize
the friction

you're getting
from other
people.
So,
it aligns
with the
concept of
sociometer
theory.
It taps
into this
notion
of cultness.
So,
if you
consider
that there's
effective
states
that enable
an individual
to track
and respond
to discrepancies
between
self-perception
and social
expectation
through these
effective states
such as
anxiety
or positive
affect,
we can see
that there's
a dynamic
nature

in self-organization
of self-presentation
of this
team,
which is
continually
shaped
and reshaped
by experiences
and interactions.
And so,
under active
inference,
we can formalize
this by saying
that a model
that is more
precise
effectively captures
the hierarchical
structure of the
moment-to-moment
variation
of more stable
traits such as
what we consider
to be self-presentation.
And so,
basically,
we can see
that identity
and group
membership
here are
fundamentally
related to
the effective
experience

of being
fit for the
group membership.
And so,
once you're there,
you can grow
in social
cohesion
and this will
lead you
to potentially
try to
maintain
the group.
It will
lead you
to try
to justify
the beliefs
of the
group
such that
you have
a form
of confirmation
bias,
but also
you maintain
the epistemic
reality of
this group,
the symbolic
reality of
this group,
which would be
more costly
to destroy
than to

actually support.
It's also
integrated into
your self-esteem,
therefore,
destroying the
group would
mean admitting
that you are
potentially not
fit for the
environment,
that you are
potentially some
degree of wrong.
self-esteem.
And so,
this leads us
to the possibility
that if someone
has low self-esteem,
what do they do?
They can either
stay in this
group and accept
that they are
unfit for the
group and
they would try
to remove
themselves to
some extent
from the
group to
avoid all
that friction
or they can
try to join

a different
group,
one which
will support
your beliefs
about the
world.
If your
beliefs are
too far
from where
the group
you're
supposed to
belong to
currently is,
and this
is where
we get
to these
sub-clusters
forming,
these little
sub-groups
that potentially
can even
become sometimes
violent and
sometimes really
not.
Sometimes it's
just, you
know,
some
tumbler group,
but it
has power
because it

gives the
individual
this feeling,
this valence
of now being
fit to a
group that
supports
their
model.
So I
wanted to
give A a
space to
talk.
Yes, thank
you.
So I would
like to go
back a bit
on what I
said about
individualism.
So yeah, I
agree that I
was wrong about
associating active
inference with
individualism because
what is interesting
here is that there
is an individuality
through active
inference which
emerge from the
collective,
active and
you can justify

or describe
individual
actions or
else through
the landscape
as you did
just now and
then go the
other way
around.
So we
here we
seem to grasp
the interaction
between the
individual and
the group
and vice
versa,
which is what
I found
interesting here.
and okay,
moreover,
there is the
what the
individual adds
to a group.
I see that
here it seems
to be captured
as we take
the individual
as a,
how to say
that,
a unique
perspective.

This is what
you said before
now,
that each
person is
a perspective,
a unique
one.
And yeah,
it's not all
what I wanted
to say.
Also,
yeah,
Lorena,
you talked
about the
beliefs that
exist.
And I think
this is one
interesting thing
also in active
inference,
it's that we
take belief
as a material
thing that
influences us
and that we
can also
influence or
create in a
way.
even,
yeah,
independently of
if it's right,

wrong,
or whatever
we judge it
is.
Like,
what we talked
about with
the moon
sect,
and the
unification
church,
yeah.
So,
there are people
with crowns,
and we say
that they are
the big
boss,
and we have
to deal
with it
or to go
away from
the landscape.
And yeah,
it was like
that in the
social world,
indifferently of
if the person
is really
the,
like,
the chief
of the cults
or not,

because they
said the
worst,
and people
were just
following them.
So,
in the end,
it happened to
create a
landscape in
which the,
in which one
person is the,
the,
the person who
has power.
So,
yeah,
I don't know
what to say
more about
all of this.
I guess I agree
with what you
said just
before.
So,
yeah.
That's it.
So,
maybe we
should move
to,
like,
challenges
and opportunities
of active

inference.
So,
two points.
I think,
like,
I just put them
out there,
you can react.
I think
that one,
active inference
may give us
computational
models,
or even,
like,
thorough-rounded
conceptual models
of social
organization,
specifically,
the,
like,
semantic
dimension
to social
life.
And I
also think
that in its
current formulation
within the
Michael
system theory,
it is
inapt to
represent
per-end evolution

because
Michael
system entail
the formulation
of a set
of states
that do not
change through
time,
and a
chemical flow
that does not
either change
through time.
so there
are opportunities
to apply
the model
to the
framework
to ground
the meaningful,
semantically,
complete models
of social
evolution,
but this
model,
if they're
comfortable
in evolution,
they will not
really vibe
well with
the math
that we
invoke
to say

our computational
models are
meaningful,
and that's
like rise
and a
rare of
questions,
I guess.
So does
anyone want
to react
to these
two statements?
Yes.
Could I make
a comment?
Just more
specifically,
I'm interested
in the ways
in which
the ideas
of active
inference
impact on
the way
we think
about society
and the way
society works.
I can see
from my reading
and from
listening to
the discussion
that there's
a very rich

new lexicon
and vocabulary
set of concepts
for the
thinking
within active
inference
thought,
and as we
just commented,
it's quite
heavily
mathematical,
so the
barrier to
entry
for the
full active
inference
worldview
is
over time
these ideas
will permeate
out into
the mainstream
and they will
change the
way people
think and
talk about
society.
I'm interested
in the
ways people
who've
studied
a bit

more
than I
have
see
potentially
how
this
will
change
the way
society
works
or the
way
that
we
have
discussions
or
envisage
society
more widely
than just
an academic
discussion
obviously.
maybe the
question's a bit
too big.
I mean,
I think
everyone is just
waiting for the
other to talk
first.
Mao talked
first.
soul.

So I
think I
want to
tackle a
little bit
what you
said about
the
open-ended
nature,
the open-ended
evolution
of
state spaces
basically.
I think
under the
path
formulation,
this
open-endedness
is actually
just a
perceived
open-endedness.
It's just
a function
of
the
timescale
being
considered
and the
motion
along
that
timescale.
I think

we can
go a
little bit
more into
depth
about
this.
I think
it's a
very large
topic.
Effectively,
if we
go into
quantum
formulation,
there's
some
degree of
irreducibility
and anything
above that
is historically
contingent.
So obviously
some things
will,
some marker
blankets will
dissolve and
reform and
dissolve and
reform.
And through
this dissolution
and reformation,
you get a
new potential

for mapping
of observation
to states.
So there's
always that
possibility.
But I
think some
challenges
for active
inference that
are coming
is, for
instance,
the
complexity
of social
behavior.
You know,
we can
extend active
inference to
capture the
nuances of
social
interactions,
but effectively,
active inference
is about
abduction.
the
best guess
for now.
And it
requires for
us to put
into the
models what

we think
are useful
variables,
what we think
are the right
variables.
variables.
And it's
possible that
we're always
going to fall
just a little
bit short.
Like, we're
always, we
can either
throw a bunch
of data at
a problem and
be like,
okay, well,
just remove
the noise,
only pick up
the signal and
figure out
what's happening,
but that's no
better than
big data and
deep learning.
So, we
have to figure
out a way
to move
forward and
understand the
variables of

interest and
do this
across scales.
So, identify
the priors that
take over at
a given moment,
but they won't
take over even
if they're higher
in the hierarchy
at another
moment because
in this
hierarchy, you've
actually switched
states, like
you've moved.
And this is only
predictable once
you've understood
the effect of
those priors
and you can't
just consider
that you have
the priors to
begin with.
So, yeah,
as far as
integration with
these existing
theories, I
think we have a
nice opportunity
ahead of us.
It's basically
what you said,

Colin.
I think we're
going to be
able to more
and more through
the alignment
of certain
kinds of
language and
the formalism
under that
language.
we're going
to be able
to pull
more and
more theories
and find
what's at
the core
of the
connection
between
these
theories.
Active
inference is
supposed to
be scale-free.
So, it
describes, or
it is possible
to help us
describe the
phenomena
basically from
sociology and
literature to

biology.
So, if we
understand this,
we understand
that there's a
possible touchpoint,
convergence of
all these
fields that
are tapping
into the
same ideas
or into
some connecting
points that
allow us to
understand how
we maybe do
away with
these different
fields, remove
the silos, and
come back
together with
something that
feels more
integrative.
But then again,
we're back to
the computational
limits, obviously.
like if I want
to compute the
prediction for
an entire
country, what
is the
granularity level

at which I
should compute
the right
variables?
And if we
understand that
individuals are
formed by the
group, etc., do
I just do a
mean field over
the entire
group?
In which case,
am I losing
out on some
of the variables
that may
actually impact
the outcomes?
I think these
are the key
questions that
remain to be
answered to
some extent.
So I agree
with what
you said
now.
I agree
that we
can't
represent
things and
somehow not
cut anything
to the

thing.
When we
model or
we represent,
the point of
it is to
have a more
compressed
way to
engage with
the thing
than the
actual
thing.
So we
will just
cut
information
that is
the point
of
modeling
or
presenting
and
maybe
in a
very
simple
system
such
like
perfect
gas
model
in
physics
there

is
information
we
can
cut
and
lose
nothing.
We
can
cut
information
about
which
particle
is
where
because
the
different
particles
of the
same
atom
has
the
same
properties
so
we
don't
really
care
what
atom
does
what
but

in
the
case
of
human
societies
we
have
patterns
of
engagement
and
the
specific
information
that is
accessed
by
a specific
agent
is
important
and
whatever
we
choose
to
cut
for
model
it
will
be
relevant
at
some
point
!

At
least
it
will
probably
be
if
the
information
we
cut
is
not
specific
to
a bunch
of
agents
that
will
die
of
car
accidents
or
something
if
any
kind
of
information
we
may
want
to
attract
out
will

be
relevant
somewhere
at
some
point
so
we
are
dealing
with
complex
systems
we
are
not
going
to
model
everything
but
it
is
a
huge
deal
that
at
the
semantic
contents
from
dynamics
either
from
the
dynamics
of

a
component
system
say
neural
dynamics
and
biological
dynamics
or
dynamics
of
a
social
context
with
the
modeling
of
action
influence
agent
as
things
that
navigate
the
world
and
make
inference
and
stuff
and
to
my
knowledge
we

do
not
have
any
other
theory
that
makes
the
bridge
between
just
dynamics
and
semantics
we
have
other
neurological
like
cognitive
neuroscience
people
who
talk
about
dynamics
a lot
and
how
we
can
bring
back
the
activity
of
cognitive

systems
to
dynamics
but
usually
it
is
in
a
way
that
insists
that
we
should
just
look
at
the
dynamics
we
don't
really
need
to
worry
about
the
semantics
or
if
we
need
to
worry
about
the
semantics

they
somehow
reduce
to
the
dynamics
the
dynamics
is
what
exists
and
is
important
and
what
we
should
worry
about
and
so
another
framework
that
is
relevant
to
like
just
the
evolution
of
norms
or
presentation
or
mental

content
in a
context
i

it

is
illusory
it
is
about
like
it
so
maybe
to
give
a bit
of
context
the
path
integral
formulation
of
the
fep
is
just
when
you
do
the
fep

so
the
bit
of
math
we
make
a
huge
deal
about
and
we
apply
it
not
to
like
the
dynamics
of
a
particle
per
see
like
how
it
affects
the
particle
at
point
t
but
how
it
affects

the
particle
for
mathematic
reasons
reframing
a problem
in
path
integral
formulation
is a
huge
deal
it
is
what
makes
the
difference
between
a
newtonian
and
full
fledged
quantum
gauge
theories
that
underwrite
all
of
physics
so
yes
path
integral

formulation
is a
huge
deal
but
to
write
it
you
still
need
to
write
an
element
space
to
make
the
list
of
all
things
that
are
possible
and
if
you
think
that
this
is a
correct
model
of
the
world

and
this
is
everything
there
is
you
have
to
agree
either
that
cause
do
not
exist
or
that
there
is
a
place
in
the
primal
web
function
that
existed
before
time
did
there
is
a
place
where
con

s
lies
and
I
think
cause
exists
and
I
do
not
think
cause
are
basic
properties
of
quantum
mechanics
so
cause
must
have
happened
along
the
way
somehow
and
to
represent
that
being
to
represent
open-endedness
in
so

we
need
open-endedness
in
math
and
the
FEP
is
not
one-ended
and
structurally
it's
very
hard
to
account
for
evolution
and
at
a
more
prosaic
level
when
we
look
at
engagement
with
social
dynamics
with
the
social
world

we
look
at
one-endness
because
there
is
always
a
integration
of
norms
there
is
always
a
loss
of
information
but
also
that
is
constructive
of
new
norms
there
is
writing
happened
at
some
point
and
then
we
did

things
differently
it
enabled
administration
to
exist
as
a
non-trivial
causal
power
society
that
actually
mattered
in
its
organization
and
new
things
happen
and
if
we
want
to
look
at
social
engagement
it
is
critical
to
look
at

how
new
things
happen
and
currently
the
FEP
doesn't
do
that
in
France
I
don't
think
it
does
that
maybe
it
can
at
a
computational
level
without
grand
pretension
maybe
but
I
did
not
see
it
and
that's

a
big
challenge
that
we
have
ahead
of
us
and
by
we
I
mean
everyone
not
just
I
can
influence
people
everyone
should
tackle
that
at
some
point
Daniel
you were
raising
your
hand
I'll
add
just
a few
notes

to
Colin's
question
certainly
it's
something
that we
all
ask
ourselves
and
many
viewers
and
participants
in
the
ecosystem
ask
like
so
what
what
then
what
does
it
really
mean
for
active
inference
to
be
applied
to
or by
or

in
the
social
sciences
so I
think
that's
a big
open
question
certainly
one
that's
not
only
addressable
by one
perspective
or even
with words
alone
but I
think
there's
a few
key
pieces
especially
that
differentiate
it from
the broader
or adjacent
project of
quantitative
social science
or formal
social science

the question
we should
seek after
is what
does active
inference
bring in
and
again
probably
many ways
to say
it but
we know
for an
active
inference
agent
as a
scale
free
statement
free energy
can be
bounded by
changing the
world or
changing the
mind
so having
continuity
of social
systems
with all
other
systems
spatial
temporal

arbitrary
material
and so
on
gives a
continuity
and an
interoperability
that without
a scale
free first
principles
approach
is essentially
not even
on the
radar
so that's
one
massive
piece
and then
I think
where that
begins to
bear
or modify
fruit
is
essentially
the
scientific
and professional
development
of cognitive
engineering
understanding
that the

cognitive
constraints
and
scaffolds
scripts
all of
these
tasks
implicitly
and then
more recently
explicitly
and most
recently
quantitatively
and formally
are examples
of cognitive
engineering
so whether
it's a
stop sign
and what
that comes
to mean
in different
social locations
or language
as a complex
phenomena
these are
examples
of cognitive
engineering
engineering
in the
same way
that

building
a stack
of rocks
is both
an example
of material
engineering
and cognitive
engineering
through semiotics
and now
when the
scientific
process
the professional
process
enters into
this reflexive
and relational
posture
through active
inference
that is
kind of a
saddle point
into a
different
dynamic
I believe
Lorena
Lorena
unmute
and then
go for it
yeah
I just
make
one comment

upon what you
said
like keep
in mind
that active
inference
can understand
beliefs
and signs
and this
kind of
semiotics
as material
and concrete
because
constraints
they constrain
the evolution
of the system
like how the
system
complexifies
right
and the
constraints
has this
material
effect
on the
dynamics
right
you cannot
really
believe that
of course
belief is
materialized
as a table

of course
not
but for the
system
in terms
of constraining
and understanding
how things
scale up
and how
to complexify
they have
that
kind of
role
and that
helps to
understand
a little bit
what Valvo
is bringing
about
how you
can bring
new
values
and parameters
for the
dynamics
that are being
described
by the
mathematics
of
external
and internal
free energy
principle

updating
right
and
that is a
way of
tackling
this
novelty
for
this
kind of
thinking
over time
but we
don't have
a proper
account
on how
to
predict
what is
going to
constrain
the
system
next
right
and
yeah
Mao
go for
it
and
then
we'll
all
kind of
give a

closing
round
slash
thoughts
okay
well
there was
a question
in the
chat
at one
point
so I just
wanted to
potentially
address the
question
if that's
okay
so the
question
was
so would
the
generative
process
of
modeling
as a
social
system
enclose
multiple
agents
on one
market
blanket
or I

assume
inside or
under
one
market
blanket
so
we
like
to
think
of
active
inferences
nested
in
hierarchical
that's
one
thing
that
Maxwell
Ramstad
definitely
pioneered
where
each
agent
has
its
own
market
blankets
but
there
is
a
higher

level
market
blanket
that
encloses
the
social
system
in
play
so
individual
agents
can
interact
with
each
other
and
the
environment
but
the
social
system
as
a
whole
also
interacts
with
the
broader
environment
and
this
is
where

semantics
and
semiotics
come
into
play
and
the
narratives
around
certain
groups
start
shaping
how
they
can
interact
with
the
other
groups
which
have
a
perception
of
these
narratives
so
there's
in-group
and
out-group
dynamics
so
it's
a

little
bit
akin
to
how
cells
within
an
organism
have
their
own
boundaries
but
are
also
part
of
a
larger
organism
so
yeah
I think
that
does that
answer
the
question
from
the
person
who
asked
that
question
I think
it does

yeah
nice
so
perhaps
A and
Colin
feel free
to give
any remarks
and then
we'll
pass
around
the
instructors
it can
be as
long
or as
short
as you
like
okay
may I
go
first
so
I would
like to
say that
the
key issue
we are
like
trying to
grasp
is that
effective

inference
has
social and
political
implications
due to
its
scale
free
structure
and how
we can
apply it
to
the
social world
then there
is the
self-referential
question that
happens which
is
what is the
structure
of
active
inference
as a
social
construction
itself
and a
political
construction
itself
and
what are
our own

like
our own
what is
our own
political
power and
social power
in this
social and
political
world
and this
is the
like
the
factal
like
thing that
is going
on like
because
afterwards
what is
my own
social and
political
implication
in this
thing
and
the more
you
you get
your own
agency
and the
agency
of active

inference
the more
it becomes
like
a lot
to process
and the
more you
you
you grasp
your own
identity
and
and the
power you
have
and the
how to say
the responsibility
yeah
you have
so yeah
I guess
this is one
of the key
points we're
trying to
to
to get
here
yeah
I guess
that's
the
the point
I wanted
to make
and

yeah
to finish
with that
we
the
the game
we are
trying to
play
is to
to try
to find
our own
blind spots
in this
game
like
what is
the limit
of the
model
and
does it
have
a
self-referential
limit
or not
so
yeah
that's it
yeah
I would
I would
endorse that
and
that was
yeah

I feel the same
way
I found that
my reading
of
just in the last
year or so
of active
inference
ideas
I find it
very empowering
the ideas
of
active
inference
in
relation
to the
individual
mind
are
quite well
developed
and
it's
yeah
very empowering
to be able
to look at
yourself
through the
active
inference
paradigm
and understand
the thought
processes

that is
you
it's a
very empowering
thing
and
as
I was
saying
just
recently
starting
to look
at the
idea
that
we can
incorporate
an active
inference
concept
of the
way
the
social
world
around
us
is
structured
which
is
part
of
that
paradigm
too
I

think

is

potentially

more

empowering

but

it's

early days

yet

but

I look

forward

very much

to

a more

developed

world

where

people have

discussions

with each

other

not just

about

what I'm

thinking

and how

I think

but how

I think

about

society

and how

we think

about

society

together

and it's

embedded
within that
paradigm
I feel
that's
potentially
something
which is
very
empowering
and I
look forward
to it
and I
thank the
speakers
all for
their
contributions
to this
series
of talks
Thank you
Colin
So
with the
exception
of
Ben
whose
lecture
and
contribution
kicked
off
this
course
and

were
excellent
also
we have
the
others
of the
instructors
here
so
how
do each
of you
want to
kind of
conclude
this
chapter
Yeah
Mal
then
Lorina
or
yes
I think
of
entities
which are
the
capability
of
interacting
to maximize
the
information
they gather
from the
environment

such that
as a
whole
they can
self-evidence
to
maintain
the
phenotypes
they believe
to be
similar
to their
own
and therefore
themselves
and how
this leads
to
over time
the
sedimentation
of
norms
and
activities
and
eventually
the
reference
to
these
norms
and
activities
through
the
semantics

and
the
scripts
and
effectively
how
this
leads
us
to
all
the
complexities
of
language
and
how
we
may
now
be
at
the
point
where
we
have
a
simulacra
that
represents
nothing
it's
just
we
just
have
these

shells
of
representations
and
these
shells
of
semantics
which
potentially
have
lost
their
anchoring
in
reality
so
I
think
that
was
the
through
line
of
the
course
I
hope
we
made
a
good
case
for
it
and
how

active
inference
allows
us
to
model
this
entire
path
not
only
across
scale
but
across
time
as
well
so
yeah
that's
how
I
think
I'm
going
to
close
yeah
I
think
this
is
me
now
yeah
I
really

like
the
story
as
well
that
we
are
building
throughout
the
whole
course
starting
with
a
very
general
idea
of
what
is
active
!
inference
with
Ben
and
then
trying
to
build
up
this
collective
notion
how
we

contain
information
in a
certain
way
that
it's
always
synchronized
so
this is
always
sort
of
meaningful
that
can be
described
in
certain
ways
by the
mathematics
of
active
inference
and
understanding
these
questions
the
hard
questions
that
we're
asking
now
like

how
we
can
bring
new
variables
how
can
bring
novelty
into
the
system
which
is
something
that
we
don't
have
an
answer
yet
but
I
find
that
it's
really
cool
that
we
don't
hold
back
in
asking
that

because
I
think
that's
the
next
step
for
understanding
this
how
complexification
comes
about
how
can
we
describe
that
Daniel
for
putting
this
together
it
was
pretty
cool
and
I
hope
you
guys
also
have
enjoyed
it
thank

you
Lorena
Afel
so
thank
you
you
welcome
I
think
the
I
put
his
finger
on
something
quite
deep
as
usual
which
is
that
if
we
want
to
have
a
coherent
theory
of
nature
that
is
not
based

on
just
here
are
the
thing
that
can
be
and
we
will
draw
dynamics
from
that
that
is
compatible
with
upper-ended
evolution
we
need
to
address
self-engeniality
we
need
to
address
like
not
consider
but
recursion
and
the

possibility
for
recursive
individuation
of
physical
beings
and
that's
tough
and
that's
very
tough
because
the
basis
of
representation
is
just
I
list
array
of
things
I
want
to
represent
and
represent
that
but
if
things
can
just

happen
that
are
new
and
that
are
by
definition
not in
the
field of
possibilities
that
pre-existed
then
we
are
fucked
there

is
no
way
we
can
represent
that
and
there
is
there
is
a
familiar
of

approach
in
quantum
information
theory
that
tries
to
tackle
that
how
to
draw
physics
without
being
rolling
in a
specific
state
space
and
just
I'd
like
to
explain
the
story
of
that
so
the
math
it
draws
from
algebraic

topology
so
we
study
surfaces
but
not
by
studying
surfaces
by
studying
transformation
within
those
surfaces
but
then
we
remove
the
surfaces
and
we
keep
only
the
transformation
over
what
I
don't
know
but
somehow
we
can
I

guess
reconstruct
coherent
mathematical
entity
from
that
and
I'd
like
us
to
take
a
look
at
how
this
look
what
the
math
look
like
!
So
can
you
can
we
see
the
screen
okay
so
this
is
the

structure
of
the
quantum
information
formulation
of
the
FEP
these
are
dimension
of
interaction
between
system
A
and
system
B
what
defines
the
scope
and
mode
of
this
interaction
I
don't
know
and
this
is
a
cocon
diagram

that
represents
an agent
together
with
the
representation
that
develops
and
I
could
tell
you
the
words
that
correspond
to
the
thing
within
this
diagram
but
I
do
not
know
what
they
mean
so
it
would
be
a
hard

thing
to
do
and
the
math
that
is
capable
to
tackle
the
self-forcibility
like
maybe
because
I
do
not
see
any
proof
that
it
actually
does
in
fact
it
is
not
tractable
for
humans
and
I
exclude
mathematical

physicists
!
of
what
is
going
on
and
because
of
the
lack
of
any
abuse
reference
that
from
anything
we
intuitively
understand
in
this
and
the
work
that
we
have
to
do
if
we
want
this
to

move
forward
and
to
have
a
self-coherent
theory
of
social
evolution
is
to
somehow
entangle
from
that
concrete
meaning
things
that
apply
to
the
world
and
conversely
use
the
way
that
we
can
think
of
it
as
concrete

interaction
in
the
world
to
further
specify
the
mathematics
and
prove
that
it
is
self-coherent
and
applies
to
stuff
and
this
is
a
big
work
in
fact
and
this
is
kind
of
exciting
I guess
what I wanted
to say
is
active

inference
is
dead
long
live
active
inference
great
well
it was
a fun
course
and
this
closing
section
and
slide
kind
of
brought
this
unity
between
Chris
Fields
his
quantum
work
and
this
social
science
which
in
the
disciplinary
paradigm

are
seen
as
distant
or
with
different
systems
of
interest
and
you
can't
get
there
from
here
but
in
a
scale
free
first
principles
paradigm
it's
not
even
just
that
we
can
hold
them
side
by
side
it's

that
they
are
more
than
juxtaposed
so
it's
been
awesome
and
it's
own
great
experience
to work
with all
of the
instructors
who
shared
their
amazing
and
youthful
and
energetic
perspectives
and
we
explored
new
format
in a
new
way
!
we're

going
to
process
the
proceedings
of
the
lectures
and
the
discussions
plus
any
final
submitted
question
and
answers
through
the
syllabus
website
bring
that
together
into
a
open
access
publication
of
some
kind
so
for
people
who
are

interested
in
that
kind
of
editorial
or
curatorial
work
the
active
inference
journal
is
for
them
and
then
in
2024
and
beyond
let's
see where
active
inference
is
see where
social
sciences
are
and
hopefully
as
people
are
interested
in

take

it

from

there

we

threw

on

many

fun

ideas

but I

guess

we'll

leave

it

till

then

to find

out

all

right

thank you

everybody

bye

thank you

thank you

Thank you.
Thank you.